

Raj Khatri

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PROFESSIONAL SUMMARY

Final-year Industrial & Production Engineering student working at the intersection of plant diagnostics and applied GenAI. At Tata Motors, diagnosed a live Body-in-White production defect by separating an upstream supplier-side cause from on-line equipment wear, using regression-based trend analysis on inspection data and confirming it by physical measurement on the shop floor. Built and deployed Source-Governed RAG, a retrieval system in which source approval status overrides semantic similarity — raising retrieval accuracy from 78% to 95% and forcing refusal rather than ungrounded generation when the corpus cannot support an answer. Reads plant process reasoning and writes Python.

EDUCATION

B.Tech, Industrial & Production Engineering <i>College of Technology, G.B. Pant University of Agriculture & Technology, Pantnagar</i>	Uttarakhand, India August 2023 – August 2027
Senior Secondary (XII), CBSE · Maheshwari Public School, Kota · 75.2%	2022
Secondary (X), CBSE · Peace Public School, Gopeshwar · 97.2%	2020

TECHNICAL SKILLS

Languages:	Python, SQL (PostgreSQL / MySQL), JavaScript, Bash
GenAI & ML:	RAG pipeline design, vector search (ChromaDB), sentence embeddings (MiniLM / ONNX), grounded generation & refusal constraints, retrieval evaluation, Scikit-learn
Backend:	FastAPI, Pydantic, SQLAlchemy, PostgreSQL, Flask, Next.js, Supabase, REST API design
Data & Tools:	Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Excel (Advanced / VBA), Power BI, Git/GitHub, Render, Vercel, Neon
Domain:	Root-cause analysis, upstream/downstream fault isolation, DFMEA, Six Sigma, statistical modeling & hypothesis testing, evaluation-set design

EXPERIENCE

Tata Motors <i>Industrial Trainee — Data Analytics</i> <i>Project: "Predictive Quality Intelligence — A Data-Driven Framework for BIW Root-Cause Diagnosis"</i> - Diagnosed a recurring Body-in-White quality failure as an upstream supplier-side dimensional defect, ruling out on-line equipment wear as the cause. - Applied linear regression with R^2 confidence scoring across ~300 inspection data points, then validated the result by physical measurement on the shop floor. - Designed a containment procedure deployable without production stoppage or vendor dependency. - Built an interactive diagnostic dashboard tracking defect trends across 10 monitored quality metrics for ongoing condition review. - Authored a 20+ section technical report benchmarking the findings against DFMEA, Six Sigma, and peer-reviewed manufacturing literature.	Pantnagar, Uttarakhand (On-site) 03 July 2026 – 03 August 2026
Voldebug Innovations Pvt. Ltd. <i>Backend Developer (Part-time)</i> - Migrating a live K–12 AI-literacy platform from browser-local persistence to a Supabase (PostgreSQL) backend — schema design, row-level security policies, and role-based authentication for student, teacher, and principal accounts. - Implementing server-side APIs in Next.js (App Router) for XP/badge progression, class-code enrollment, and teacher analytics, replacing client-only state with a shared data layer. - Establishing database migrations, staging/production environment separation, and a repeatable deployment workflow on Vercel.	Remote 22 August 2026 – Present
Intern - Built the platform's first working version — a gamified AI-literacy SPA spanning student, teacher, and principal roles, with a curriculum-linked prompt library and an XP/badge/streak progression system. - Developed AI-assisted lesson-planning tools and a principal-facing analytics dashboard; implemented Hindi/English bilingual support and downloadable student certificates. - Authored the phased Supabase + Next.js migration plan now being implemented in the backend role above.	01 July 2026 – 31 July 2026

PROJECTS

Source-Governed RAG <i>Python, FastAPI, ChromaDB, MiniLM/ONNX, PostgreSQL, Next.js · Render / Vercel / Neon</i>	Live demo
- Built an end-to-end retrieval-augmented generation system over regulatory documents, combining vector search, grounded generation, and a governance layer that vetoes untrustworthy sources. - Improved retrieval accuracy from 78% to 95% on a labelled evaluation set by replacing fixed-size character chunking with document-structure chunking. - Architected a governance layer where document approval status and review dates override similarity ranking, preventing answers drawn from expired or unapproved sources. - Engineered grounding constraints that make the model refuse rather than answer from training data, verified by asking a question whose answer is widely known but absent from the corpus. - Swept every similarity threshold from 0.80 to 1.50 in a calibration script, measured a 73.9% accuracy ceiling, and removed distance-based confidence scoring as insufficiently reliable.	

CERTIFICATIONS

Forage Job Simulations: J.P. Morgan (Quantitative Research) · Deloitte Australia (Data Analytics) · Mastercard (Cybersecurity)	2026
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